AGRIMA DEEDWANIA

DPhil Student Interdisciplinary Bioscience, BBSRC DTP
B.Tech, Biochemical Engg. and Biotechnology, IIT Delhi
Biotech Junior Research Scientist, Inito

Email: agrimadeedwania321@gmail.com
Mobile no: (+44)7596050281, (+91)946187133
LinkedIn, Webpage

EDUCATION

University of Oxford *September 2023- Present*
DPhil Interdisciplinary Bioscience, BBSRC DTP

Indian Institute of Technology Delhi *July 2018 - May 2022*
B.Tech. Biochemical Engineering and Biotechnology Overall GPA: 8.362/10

PUBLICATIONS

- **Deedwania A**, Wang Y, Robinson CV, Bolla JR. Native mass spectrometry of membrane proteins reconstituted in peptidiscs. RSC Chem Biol. 2025;7(1):120-128. Published 2025 Oct 22. doi:10.1039/d5cb00236b
- **Deedwania A**, Karmakar S, Kumar V, Shefrin S, Sundar D, Srivastava P. Construction and characterization of a temperature-sensitive pRC4 replicon for Rhodococcus and Gordonia. Gene. 2024 Feb 20;896:147990. doi: 10.1016/j.gene.2023.147990. Epub 2023 Nov 15. PMID: 37977321.

PROJECTS AND RESEARCH

Single-Cell Insights into *Bacillus subtilis* Resistance to Antimicrobial Peptides: (dry, wet lab)
Guide: Dr. Harrison Steel, Dr. Jani Reddy Bolla (University of Oxford) *(August 2024- Present)*

- Performed bulk measurements for **growth kinetics and MIC assays** using plate-reader experiments to quantify bacterial susceptibility to nisin across wild-type and engineered strains.
- Optimized **PDMS fabrication and plasma bonding protocols** to reliably produce mother-machine microfluidic devices with channel features as small as **300 nm** while minimizing structural collapse.
- Established a **bubble-free microfluidic workflow** for long-term single-cell experiments by integrating **valve-based chemical switching and a recirculation system** for controlled cell injection.
- Conducted automated **time-lapse fluorescence microscopy** to capture single-cell growth dynamics, antimicrobial susceptibility, and membrane surface charge at single-cell level.
- Developed a **Python-based image analysis pipeline** using **DeLTA segmentation** for single-cell tracking, growth curve extraction, and fluorescence-based surface charge quantification.

Native mass spectrometry of membrane proteins reconstituted in peptidiscs [1]: (wet lab)
Guide: Dr. Jani Reddy Bolla (Department of Biology, University of Oxford) *(January 2024- July 2024)*

- Optimized **membrane protein purification** and **peptidisc reconstitution** assembly to preserve native protein complexes for structural characterization.
- Compared reconstitution strategies using **on-column and on-bead** assembly methods to evaluate peptidisc compatibility with native MS for membrane protein structural studies.
- Conducted **native mass spectrometry** to characterize protein complex integrity, oligomeric state, and stability following peptidisc reconstitution.

Construction of Temperature Sensitive Plasmid Replicon for *Gordonia* [2]: (dry, wet lab)
Guide: Prof Preeti Srivastava (DBEB, IIT Delhi) *(March 2021-August 2022)*

- Predicted the **temperature-sensitive mutant** of pRC4 plasmid replicon using bioinformatics tools.
- Performed **homology modeling** and **DNA-Protein docking** to predict and compare interactions.
- Performed **mutagenesis** of WT repB protein and screened for TS phenotype in *Gordonia* and *Rhodococcus*.

[1] Publication: https://doi.org/10.1039/d5cb00236b
[2] Publication: https://doi.org/10.1016/j.gene.2023.147990

Creating Large Protein Libraries: An Efficient Engineering Approach: (wet lab)

Guide: Dr. Harrison Steel (Department of Engineering, University of Oxford) *(April 2024- June 2024)*

- Developed and optimized a methodology for **large (targeted/non-targeted) mutant library** generation using **SLUPT mutagenesis** and generated a protein library for GFP variants.
- Performed **high-throughput screening of GFP variants** using plate-reader excitation/emission spectroscopy to identify spectral shifts and characterize mutants with altered fluorescence properties.

ssDNA Export from Synthetic Cell and Linker Induced Fusion of Cells[3]: (dry lab)

Guide: Prof Richard M. Murray (CDS and Bioengineering, Caltech) *(May 2020-August 2020)*

- **Designed mathematical models** for import/ export of molecules in a synthetic cell.
- Designed an efficient protocol for **inducible gene expression** of the membrane transport channel.
- Designed protocol for **export of ssDNA** and **fusion of two cells** induced by the exported DNA.

RNA Thermometers: (dry, wet lab)

Guide: Prof Shaunak Sen (Department of Electrical Engineering IIT Delhi) *(Aug 2019- December 2020)*

- Designed **nucleic acid structures and melt profile** of RNA thermometers at different temperature.
- Experiments of protein expression of thermometers like rpoH, agsA, cspA at different temperatures.

INDUSTRY EXPERIENCE

Inito[4] *June 2022- July 2023*

Biotech Junior Research Scientist

- Worked on **Lateral Flow Blood Assays**, primarily solved the issue of RBC interference in LFAs. Developed assays for thyroid and testosterone hormone measurement from blood samples.
- Established an antibody production system and **engineered antibodies** for LFAs using bioinformatics tools to **reduce the cost and increase the sensitivity and stability** of the existing product.

SKILLS AND STRENGTHS

- **Molecular Biology & Microbiology:** Microbial cell culture, gene knockout and complementation, molecular cloning, site-directed and random mutagenesis, PCR, DNA/RNA extraction and purification, bacterial transformation (chemical and electrocompetent), mutant library generation, protein expression, growth kinetics, MIC determination, ELISA, lateral flow assay development, plate reader-based assays, bioreactor design and operation for microbial cultivation.
- **Protein Biochemistry & Structural Biology:** Membrane protein purification, AKTA-based protein purification (FPLC), native mass spectrometry, protein engineering and production, fluorescence spectroscopy.
- **Microfluidics & Microscopy:** PDMS fabrication, plasma bonding, mother-machine microfluidics, valve-controlled fluidics, fluorescence microscopy, automated time-lapse imaging, single-cell phenotypic analysis.
- **Computational Biology & Data Analysis:** Software automation, DeLTA-based single-cell segmentation and tracking, quantitative image processing and analysis, fluorescence quantification, growth curve analysis, bioinformatics, homology modelling, molecular docking, mathematical modelling.
- **Programming & Software:** Python (scientific computing, image analysis, data processing, automation), C/C++, ImageJ/Fiji, SnapGene, Benchling, Microsoft Office Suite.

AWARDS

- **Awarded a fully funded UKRI-BBSRC Studentship** for Interdisciplinary Bioscience Doctoral Training Partnership to pursue a DPhil at the University of Oxford (2023-2027).
- **Best Academic Poster Presentation Award** at the Doctoral Training Partnership (DTP) Annual Symposium, 2026, for research on single-cell antimicrobial peptide resistance.
- **Gold Medal Winner** at iGEM 2022 for the lead recovery project[5]. **Advisor** for the IIT-Delhi team.

[3] Jupyter notebooks for mathematical models: https://github.com/agrimadeedwania/DNAexp_vesiclefusion.git

[4] https://www.inito.com/

[5] https://2022.igem.wiki/iit-delhi/